

Groups of 7

The algorithm will work if the elements are divided in groups of 7. On each partitioning, the minimum number of elements that are less than (or greater than) x will be:

$$4\left(\left\lceil\left\lceil\frac{1}{2}\left\lceil\frac{n}{7}\right\rceil\right\rceil-2\right)\right)\geq\frac{2n}{7}-8$$

The partitioning will reduce the subproblem to size at most $5n/7 + 8$. This yields the following recurrence:

$$T(n)=\begin{cases}\mathcal{O}(1) & \text{if } n < n_0 \\ T(\lceil n/7\rceil)+T(5n/7+8)+\mathcal{O}(n) & \text{if } n \geq n_0\end{cases}$$

We guess $T(n) \leq cn$ and bound the non-recursive term with an :

$$\begin{aligned}T(n) &\leq c\lceil n/7\rceil + c(5n/7 + 8) + an \\ &\leq cn/7 + c + 5cn/7 + 8c + an \\ &= 6cn/7 + 9c + an \\ &= cn + (-cn/7 + 9c + an) \\ &\leq cn \\ &= \mathcal{O}(n)\end{aligned}$$

The last step holds when $(-cn/7 + 9c + an) \leq 0$. That is:

$$-cn/7 + 9c + an \leq 0$$

$$\Downarrow$$

$$c(n/7 - 9) \geq an$$

$$\Downarrow$$

$$\frac{c(n-63)}{7} \geq an$$

$$\Downarrow$$

$$c \geq \frac{7an}{n-63}$$

By picking $n_0 = 126$ and $n \leq n_0$, we get that $n/(n-63) \leq 2$. Then we just need $c \geq 14a$.

4. (CLRS 9.3-7) Describe an $O(n)$ algorithm that, given a set S of n distinct numbers and a positive integer $k \leq n$, determines the k numbers in S that are closest to the median of S .

Solution: Assume for simplicity that n is odd and k is even. If the set S was in sorted order, the median is in position $n/2$ and the k numbers in S that are closest to the median are in positions $(n - k)/2$ through $(n + k)/2$. We first use linear time selection to find the $(n - k)/2$, $n/2$, and $(n + k)/2$ elements and then pass through the set S to find the numbers less than the $(n + k)/2$ element, greater than the $(n - k)/2$ element, and not equal to the $n/2$ element. The algorithm takes $O(n)$ time as we use linear time selection exactly three times and traverse the n numbers in S once.